Round 4: weighted grading (G vs F), the swap-symmetry split, the

crossover with rho, and the big-host spec 2026-11-04 · follow-up to `../round3/README.md` (#3380). The four items were chosen and their expected outcomes stated in chat before any run. That is not a formal pre-registration, which this round does not have. · `WEIGHTED.md` (item 2) · `parity.sing` (item 4) · `CROSSOVER.md` (item 3) · `BIGHOST.md` (item 1) · logs `logs/` · graph `round4.svg` · PDF `report.pdf`

Class: stage diagnostics plus a derived bound. No pipeline is priced and no leaderboard row changes.

Result

1. **Item 2: weights do not rescue the glued arm G.** In Singular 4.3.2 `std` at m = 4, t = 4, F finished in 105 s with standard grading (`dp`) and in 97 s with its natural weights ($e_i \mapsto i$). G did not finish within 1,800 s under either `dp` or its natural weights ($s \mapsto 1$, $q \mapsto 2$). So G/F is above 17 under `dp` and above 18 under weights, both lower bounds. At the cut-off G's weighted degree (37) already exceeded F's entire run (34). G's round-3 handicap is not just a grading artefact, at least within this budget. Singular is about 10× slower than msolve on F (105 s against 10.7 s), so only the ratios matter.

2. **Item 4: the swap-symmetry split is real but buys nothing in msolve.** Exchanging the two cherries is a symmetry of the m = 4 tree. In the coordinates $a = v_0 + v_1$, $b = v_0 - v_1$, all 19 generators are even or odd in the 9 b-variables (10 even, 9 odd, none mixed); `parity.sing` checks this.

   - msolve on the adapted system gives the same D = 7 and the same quotient dimension (6), in 680 s against T's 697 s. Its widest matrix is 3.7× wider (94,422 against 25,742 columns), because the new coordinates make the polynomials denser.
   - An engine that used the even/odd block split could save at most about $2^{\omega-1}$, roughly 2–3.5×, in linear algebra. That would first have to pay back the 3.7× fill. It is a constant factor in any case.

3. **Item 3: the cost law puts the crossover with rho far beyond cryptographic sizes** (`CROSSOVER.md`). The model takes the measured degree laws ($D = 4 + 1.5(m-2)$, plus $\beta(t-4)$ with $\beta \in \{0, 0.5\}$) into the dense Macaulay cost $\omega \cdot \log_2 C(N+D, D)$.

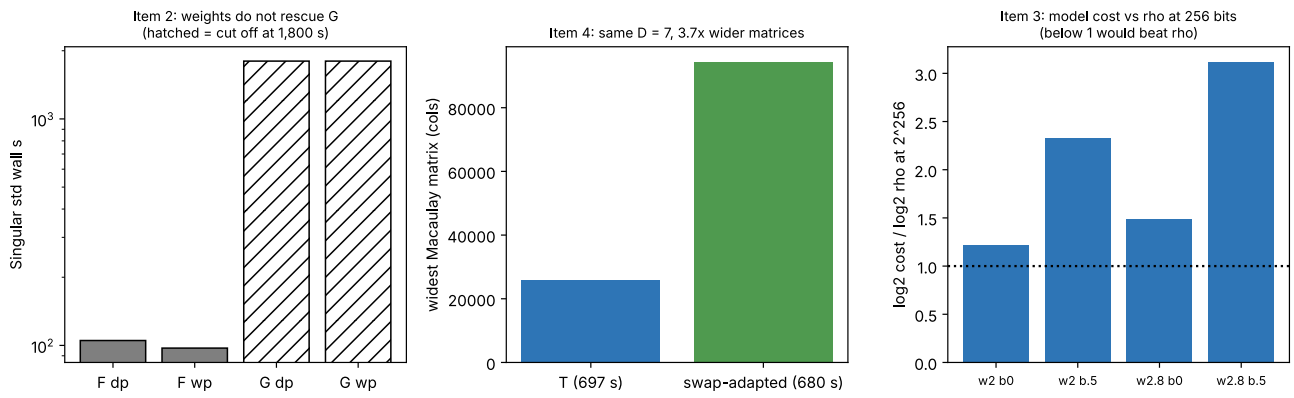
   - **Asymptotically** the model is subexponential: the ratio to $\sqrt{p}$ falls without bound.
   - **At 256 bits** even the most optimistic row ($\omega = 2$, $\beta = 0$) costs 2^{156} against rho's 2^{128}. With the measured depth slope it is 2^{298}–2^{399}.
   - **The crossover** lies roughly between 2^{500} and 2^{6000}.

4. **Item 1: the big-host run is blocked, and its specification is frozen** (`BIGHOST.md`). Two things block it:

   - This container's AWS keys are rejected by STS (`InvalidClientTokenId`).
   - The portable `isolab.eks.request/v1` contract runs only ecbench specs of at most 3,600 s, while this workload is msolve. It needs the EKS MCP's `isolab_k8s_run` or an ecbench extension.

   Inputs, resources, stop conditions and the outcome table are all fixed. None of its outcomes changes the 256-bit comparison in item 3.

Reading. All four items point the same way. The tree encoding is the only one of the measured encodings that scales in arity (rounds 2–3), and round 4 finds no encoding trick that lowers its cost: no weighted grading, no elimination, no symmetry split. Its implied cost law could be subexponential, but it would beat rho only far beyond cryptographic sizes. That is the measured distance remaining for this line. Closing it needs a PDP solver whose cost is not the dense Macaulay size, which is a different research direction from factor-base symmetry.



Records

item	record	key value

2	`logs/weighted/{F,G}_{dp,wp}.log`	F 105 / 97 s; G > 1,800 s under both orders (exit 124)
4	`logs/parity/m4_t4_k0_P.log` and `SYSTEM.sha256`	D = 7, quotient dimension 6, 680 s, widest 94,422
3	`CROSSOVER.md` §4 (awk commands and outputs)	ratio at 2^256: 1.22–3.12 across four model rows
1	`BIGHOST.md`	not run

Inputs:

- Item 2 uses the round-2 F and round-3 G systems, hashed in `../round2/SYSTEMS.sha256` and `../round3/SYSTEMS.sha256`.
- Item 4's input is the round-1 `m4_t4_k0_T` system, regenerated by `../gen.sing`.

Caveats

- **Item 2:** Singular runs used one pinned core and a 3 GiB cap each. An msolve job (item 4) shared the host during the G runs. G's runs are timeouts, not completions.
- **Item 4:** wall times are from runs that shared the host. D, quotient dimension and widths are deterministic.
- **Item 3:** a model whose degree laws extrapolate lower bounds at m = 5, 6 and $t \le 6$, with $\omega \in \{2, 2.8\}$. It is a bound, not a measured cost.

Item 3: the cost law the tree encoding implies, and where it would

cross rho 2026-11-04 · theory note built on the measurements of rounds 1–3. **Class: derived bound, not a measurement.** No result here is an algorithm cost; the model is the standard dense Macaulay-matrix heuristic, and its inputs are extrapolations from small cells, several of which are lower bounds. The numbers are computed with the `awk` one-liners in §4 (no Python).

1. Which encoding can scale at all

encoding	input degree	D measured (t = 4–5)	m ≥ 5
F (S_{m+1} in e_1..e_m)	2^{m−1} (2, 4, 8, 16, 32 at m = 2..6)	5–6 (m=3), 8 (m=4)	S_6 and S_7 were never built in the lab, and D ≥ 16 at m = 5
C (caterpillar)	≤ 4	≥ 7 at m = 4	bounded input, no symmetry quotient
T (cherry tree)	≤ 4	4, 6, 7, ≥ 9, ≥ 10 at m = 2..6	bounded input; the only measured encoding that runs at m = 5, 6
G (glued: resultants of T)	16 at m = 4	≥ 9 at m = 4	dominated (round 3)

At m ≥ 5, F's input degree alone (16, 32) exceeds T's measured D (≥ 9, ≥ 10) already. So for m ≥ 5 the tree is the only encoding left, unless a weighted grading rehabilitates the symmetric ones (item 2, `WEIGHTED.md`).

2. Model

Factor base V = g·μ_{2^t}, with about 2^{t−1} usable points; arity m; p ≈ 2^{mt} (the balanced base, one decomposition per target on average). Per relation, with $\log_2$ throughout:

- PDP solve ≈ ω · $\log_2$ C(N + D, D), with N ≈ m(t + 1) unknowns (the measured N is 4t + 2 at m = 4, 5t + 3 at m = 5);
- targets per relation ≈ m!, with the symmetry quotient charged as in the lab's convention;
- relations ≈ |F| ≈ 2^{t−1};
- linear algebra ≈ 2(t − 1), from a sparse $|F|^2$ solve.

The degree law has two parts, and both are extrapolated:

- **Arity:** D = 4 + 1.5(m − 2). This is the round-3 lower-bound slope; the measured points are 4, 6, 7 at m = 2, 3, 4.
- **Depth:** + β(t − 4). Here β = 0 is the optimistic case. β ≈ 0.5 matches T's staircase at m = 3 (6, 6, ≥ 7 at t = 4, 5, 6).

The total is the larger of (relations × PDP) and linear algebra, minimized over m. "Ratio" below is $\log_2$ cost / $\log_2$ √p, so a ratio below 1 beats Pollard rho.

3. Result

ω	β	2^256	2^1024	2^4096	2^16384
2.0	0	1.22	0.66	0.35	0.18
2.0	0.5	2.33	1.42	0.84	0.48
2.8	0	1.49	0.78	0.41	0.22
2.8	0.5	3.12	1.91	1.12	0.65

At ω = 2.8, β = 0 the optimum runs m = 4 at 256 bits (cost 2^190.8 against rho's 2^128), m = 6 at 1,024 bits and m = 10 at 4,096 bits; full table in §4.

- **Asymptotically**, under these laws, the ratio falls without bound, so the model is subexponential, as THEORY.md §3 predicted for D = O(m). This holds only if the laws extrapolate. They rest on lower bounds at m = 5, 6 and on t ≤ 6.
- **At cryptographic size it does not come close.** At 2^256, even the most optimistic row (ω = 2, β = 0) costs 2^{156} against rho's 2^{128}. With the measured depth slope it is 2^{298}–2^{399}. The crossover with rho lies roughly between 2^{500} and 2^{6000}, depending on the row (interpolated between the table columns).
- **What would change this:** a D law with a slope well below 1 per summand and β = 0. Or a PDP solver whose cost is not governed by the dense Macaulay size C(N+D, D), for example one that exploits the tree's sparsity or treewidth. The lab's SAT/treewidth leads found no exponent movement (`research/ic_prime_eth_leads_20261031/SYNTHESIS.md`).

Consequence for the other items. Confirming P2 on a big host (item 1) can only make the asymptotic claim stronger; it cannot close the gap at 256 bits. A swap-symmetry block split (item 4) is worth at most 2^{ω−1}, under 2 bits, against a gap of at least 28 bits even in the best row.

4. Commands

The best-m search with $\omega = 2.8$ and $\beta = 0$:

```
awk 'function lf(n,  s,i){s=0; for(i=2;i<=n;i++)s+=log(i); return s/log(2)} function lb(n,k){return lf(n)-
lf(k)-lf(n-k)}
BEGIN{ for (bits=256; bits<=65536; bits*=2) { best=1e12;
  for (m=2;m<=400;m++){ t=int(bits/m+0.999); if(t<2)continue; D=int(4+1.5*(m-2)+0.5); N=m*(t+1);
c=(t-1)+lf(m)+2.8*lb(N+D,D); la=2*(t-1); c2=(la>c)?la:c;
    if(c2<best){best=c2;bm=m;bt=t;bD=D}}
  printf "2^%-6d best m=%-3d t=%-5d D=%-4d log2 cost=%8.1f  log2 rho=%8.1f  ratio=%.3f\n",
bits,bm,bt,bD,best,bits/2, best/(bits/2) } }'
```

Output (recorded):

```
2^256    best m=4   t=64    D=7    log2 cost=   190.8  log2 rho=   128.0  ratio=1.491
2^512    best m=4   t=128   D=7    log2 cost=   274.0  log2 rho=   256.0  ratio=1.070
2^1024   best m=6   t=171   D=10   log2 cost=   399.0  log2 rho=   512.0  ratio=0.779
2^2048   best m=8   t=256   D=13   log2 cost=   580.0  log2 rho=  1024.0  ratio=0.566
2^4096   best m=10  t=410   D=16   log2 cost=   844.8  log2 rho=  2048.0  ratio=0.413
2^8192   best m=14  t=586   D=22   log2 cost=  1226.7  log2 rho=  4096.0  ratio=0.299
2^16384  best m=20  t=820   D=31   log2 cost=  1780.2  log2 rho=  8192.0  ratio=0.217
2^32768  best m=26  t=1261  D=40   log2 cost=  2583.1  log2 rho= 16384.0  ratio=0.158
2^65536  best m=38  t=1725  D=58   log2 cost=  3742.3  log2 rho= 32768.0  ratio=0.114
```

The table in §3 uses the same model with `W` (ω) and `beta` as variables, and `t ≥ 4`.